Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/04/2005
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
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Flight Program:			
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COI Name (Institution):	Russel, William B. (Princeton)		
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Task Description:	This experiment consists of microscopy, microscopic and mesoscopic manipulation and control of colloidal samples to probe the essential features of the hard sphere disorder-order transition and the properties of the ordered solid phase that results. Many samples, in custom-designed microscope slide style cells that permit many different volume fractions and the application of external fields such as temperature gradients, periodic pattern imposition, and force application through the use of optical tweezers, will permit measurements from the disordered fluid phase, through the coexistence region, into the fully crystalline solid, and beyond into a glass-like phase. The dispersions will consist of colloidal hard spheres, in the 0.65 to 1.0 micron diameter range, with controlled polydispersity, consisting of poly (methyl methacrylate) (PMMA) in an index-matched fluid. Microscopy will allow for the observation of the position of individual colloidal particles and thus the determination of local crystalline or liquid structure, defect structure, observation of nucleation and growth of individual crystallites, the spontaneous formation of dendrites, the effect of periodic and other imposed patterns, field induced local melting and freezing, and the dynamics of liquid-crystal, liquid-glass and solid-glass interfaces. The applied fields will allow for controlled nucleation and growth, induced metastable phases, susceptibility, and rheology measurements. A simple design with many 50 microliter samples in a single sample holder on the microscope platen will allow interchange between the many samples to be studied. Static light scattering and confocal microscopy (replacing dynamic light scattering) will be done on the same samples. The static scattering image will be acquired with a CCD for Bragg scattering and structural studies. Confocal microscopy will allow the 3-d study of local affects. We expect to determine the volume fractions at which a fluid changes to a crystal, find the crystal structure of the equilibrium solid phase, contrast nucleation and growth in gravity and microgravity, especially in the region of the ground based glass transition (which we found does not occur in microgravity), quantitatively test the dendritic growth model, explore the role of polydispersity on the phase diagram and the structure of the different phases and explore the susceptibilities to external fields and the microrheology of the different phases. Characterization of crystal formation in the microgravity environment can lead to a greater understanding of how gravity affects many kinds of colloidal materials. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices. The experiment employs a material that assumes multiple phases, which depend only on concentration. The simplest colloidal dispersion consists of hard spheres, which interact through hydrodynamic and Brownian forces, but feel no direct force before an infinite repulsion at contact. The hard sphere phase diagram, including the disorder-order transition, has been well established, but many-body interactions complicate the understanding of the kinetics of the transformations and the mechanical properties of disordered/metastable fluids and the ordered solid.Our experiments on these systems in microgravity, where the effects of sedimentation and convection are absent, have yielded several pleasant surprises. These included crystals (that were glasses on earth) and dendritic growth. In addition, Bragg images during nucleation and growth confirm some anticipated but previously unobserved features of the crystal growth process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	

Princeton University Quarterly Report Progress: (11/1/04-01/31/05)
Lattice microrheology (M. Sullivan)

We are studying the viscoelastic properties of colloidal hard sphere crystals using particle tracking microrheology. Unlike active techniques that measure the response of a probe particle to an external driving force, this passive measurement uses the Brownian dynamics of the colloidal particles to probe the rheology and structure of the crystal. Fundamentally, the rheological properties are determined solely by thermal energy, kT. The stochastic energy of the embedded, micron-sized spheres can be related to the frequency dependent rheology of the surrounding viscoelastic material [see Gardel et al., pp 1–55, in Microscale Diagnostic Techniques, K. Breuer, ed., Springer-Verlag (2002)]. This methodology is based on the fluctuation-dissipation theorem that states that the correlation of thermally-driven fluctuations is directly related to the equilibrium relaxation. Thus, by measuring particle position correlations as a function of time and using a generalized Stokes-Einstein relation, we can measure viscoelastic relaxation in the hard sphere crystal. Correlations between one particle and itself over time (single-point) as well as the cross-correlated motion of pairs of particles (two-point) are used. The advantage of two-particle microrheology is that it determines the bulk rheology rather than the structure and mechanics of the local microenvironment. This feature allows the possibility of measuring the anisotropy of the FCC lattice as a function of lattice direction.

The resolution of detecting particle centers determines the range of moduli that can be measured. A simple scaling analysis equating the thermal energy density of a constituent sphere to the elastic energy needed to deform the surrounding material a given length indicates that the detection scheme is critical. To illustrate this point, images of a hard sphere crystal comprised of 1 micron diameter spheres were obtained with the Nipkow spinning disk confocal microscope. Particle centers detected using a modified 'Crocker & Grier' subpixel algorithm [Crocker, J.C. and Grier, D.G., J. Colloid Interface Sci. 179, 298 (1996)] could be resolved to ~ 20 nm which would allow measurements of samples with an elastic modulus of ~ 20 Pa or less. We expect G'a^3/kT ~ O(10) at volume fractions in the range 0.50–0.52, indicating that our confocal microscope is capable of measuring the desired relaxation spectra.

In our experiment, a static surface template was used to nucleate and grow FCC 100 crystals in an approximately density matched suspension of PMMA-PHSA particles. The solvent mixture is comprised of tetrachloroethylene and decahydronaphthalene (TCE/decalin). Using confocal microscopy, we track particle positions as a function of time. The spatial and temporal particle correlations, e.g., the ensemble averaged mean-squared displacement, relate to the frequency dependent complex elasticity of the medium and are being used to measure the viscoelastic properties. The overall magnitude of these effects is small, however, requiring the continuing resolution of several image and analysis issues.

The heterogeneously nucleated crystals are grown with the (100) face perpendicular to the optical path (the z-axis). With the high z-resolution of the confocal microscope, it is possible to observe one single FCC (100) layer and track the motion of the particles in this layer. Using the images, particles can be tracked to subpixel accuracy over the span of the experiment. Hard sphere crystals have anisotropic elasticity so the full elastic tensor needs to be determined. Note that the FCC lattice has three independent elastic constants as opposed to two in an isotropic system. The speed of the measurement is particularly important for understanding the suspension properties– at short times the behavior is essentially viscous (i.e. the short time self-diffusion)– and initial measurements confirm this behavior. At long times the particles are confined to their particular cages, which is a reflection of the bulk elasticity of the crystal.

‘Electric bottle’ glasses (M. Sullivan, K. Zhao)

The gradient of an applied electric field across a suspension of dielectric particles, e.g., PMMA-PHSA spheres, will cause particle migration known as dielectrophoresis. The strength of the dielectrophoretic force is proportional to the gradient of the field squared and the dielectric contrast between the particles and their surrounding medium. In a closed system, this phenomenon will lead to particle concentration gradients. When the system reaches equilibrium, the corresponding osmotic pressure gradients– which depend on concentration– will exactly balance the dielectrophoretic force. Thus the measurement of the spatially dependent concentration, together with the known electric field profile, allows the osmotic pressure of the suspension to be determined as a function of particle volume fraction. We use confocal microscopy to measure the concentration profiles by tracking and counting colloidal particles across the sample.

Using dielectrophoretic equilibrium in both wedge and “bottle” geometries, we have been able to produce kinetically arrested colloidal suspensions. The slow densification rate has caused our low polydispersity samples to crystallize and then densify, precluding the formation of glasses. To inhibit crystal formation, binary mixtures of spheres (R1:R2 = 1:1.12) were prepared and used to produce colloidal glasses in about one week. The suspensions appear completely jammed (no detectable motion) with colloidal regions in coexistence with a dilute liquid phase. Particle tracking in these high density systems has proven difficult, even with confocal microscopy. To improve the image analysis, samples with only a portion of the total particles fluorescently-labeled have been prepared and are currently being equilibrated.

To determine the properties of the glass, we are using the passive microrheology techniques developed for our FCC suspensions. In addition to studying colloidal glasses, we continue to work on thermodynamically equilibrated systems and are using this technique to measure the hard sphere equation of state. The results agree well through the liquid/crystal transition (phi ~ 0.5), but we continue our effort to correctly measure and understand the results in the significantly denser crystal.

Controllable 2-d potentials (M. Sullivan)

A dipole-dipole interaction is spatially anisotropic, but the total interaction integrated over space is zero. The average over the plane of the electric field, however, yields an attractive interaction that decays as r^-3. To create such an interaction, we suspend charged PMMA-PHSA colloids in a cyclohexyl bromide-decahydronaphthalene (CHB/decalin) solvent mixture using a confined (monolayer) cell geometry. A set of four electrodes are located in plane in a rectangular configuration. These electrodes are driven 90 degrees out of phase to produce a rapidly rotating electric field in the central region. To date, we have created controllable dipole-dipole chaining and are working on producing and detecting a net attraction in the sample.

Template directed heterogeneous nucleation (M. Sullivan)

In thermodynamic equilibrium, hard sphere suspensions above a volume fraction of 0.494 are at least partially crystalline, but can be homogenized to a fluid phase using shear. Upon cessation of the applied force, a crystal structure can form from a dense metastable fluid in two distinct ways– homogenous and heterogeneous nucleation. Homogeneous nucleation occurs in the bulk when a randomly formed seed reaches a critical size and grows through the fluid. Heterogeneous nucleation occurs on some defect or bounding surface causes the crystal to form and grow in to the bulk. By controlling the surface of a colloidal suspension, we are able to grow particular crystal face in to the bulk. We modulate the surface energy in two ways– either by using a static physical template consisting of holes or by using electric field gradients and dielectrophoresis to pull particles in to a particular configuration. This will control the positions of the first particle layer, which will then grow in to the bulk. Using confocal microscopy we are able to track this crystal growth in three dimensions as a function of time. We are also able to measure the three dimensional structure of the resulting crystal and observe the effects of lattice-size mismatch and strain.

Particle characterization (A.D. Hollingsworth)

Samples of PMMA-PHSA particles prepared for Wes Kopacka’s research were submitted to Dan Phifer at FEI Company’s R&D Center (Peabody, MA) for SEM imaging. Several high resolution micrographs of iridium-coated (and uncoated) particles were obtained using a Sirion 200 Schottky field emission scanning electron microscope (SEM), as well as a Quanta 200 FEG low vacuum SEM. The objective was to compare these images with those obtained here at Princeton University’s Image and Analysis Center (IAC) using a Philips XL30 FEG-SEM at 5 kV beam intensity. We expect that using an environmental SEM or an SEM with in-lens detection will minimize artifacts such as particle shrinkage and image distortion, thus improving the determination of average particle size and size distribution. During this time period we also recalibrated the XL30 FEG-SEM magnification in response to a significant discrepancy detected between our SEM and TEM micrographs. Image analysis comparing all results (IAC-SEM, IAC-TEM, and FEI-SEM) is in progress.

Ongoing collaboration with Utrecht University (A.D. Hollingsworth, M. Leunissen)

In a low polar environment, sterically-stabilized poly(methyl methacrylate) spheres become positively charged and exhibit significant long-range repulsive interactions. Particles were fluorescently labeled for confocal microscopy and suspended in near index and density matching solvents. Calculations show that small differences in the dielectric constant, which ranged from 6 to 8 in our experiments, can dramatically affect electrolyte dissociation. We have computed ionic strengths from conductivity measurements using both Bjerrum’s and Fuoss’ theories of ionic association and, in turn, interpreted electrophoretic mobility and dielectric response of dilute suspensions using the standard electrokinetic model. We hypothesize that the dehydrohalogenation of cyclohexyl bromide (CHB) produces hydrogen bromide (HBr), which weakly dissociates in these organic solvents. To establish the acidic nature of the media, we dissolved CHB, along with its decomposition products, in water and measured the conductivity, pH, and pAg as a function of CHB concentration and time. HBr is a strong acid in water (pKa = –4.7 at 25 deg. C), while CHB is sparingly soluble: 640 microM at 18 deg. C. Note that silver nitrate is an indicator for bromide ions.

We continue our study of charged colloids near an oil-water interface. These systems produce ‘colossal crystals’ and ‘colloidosomes’ which we have imaged with confocal microscopy. In the CHB/decalin mixture, sterically stabilized PMMA spheres have a positive charge. Together with a Debye screening length of several microns, this results in long-ranged repulsive interactions. Water forms a stable interface with CHB and, with its high dielectric constant, absorbs ions from the oil phase. This behavior reduces the screening further and leads to ‘Coulombic’ crystals with extreme lattice constants (tens of microns). The oil-water interface often is charged, repelling the similarly charged particles in the oil phase and leaving a large depletion zone (~ 50 microns). However, independent of the sign or magnitude of interfacial charges, at short range the attraction between the charged particle and its image (in the water)

dominates. Thus, the charged hydrophobic particles get permanently bound to the interface, forming a 2D ‘surface crystal’. The crystal density can be controlled with an electric field that drives the particles to the surface. Using the specific properties of this particle-oil-water system we can make particle-coated droplets, i.e., colloidosomes, geometric particle packings enveloped by a water sheath, and particle transporting channels.

Princeton University Quarterly Report Progress: (08/1/04-10/31/04)

Preparation and characterization of colloidal ellipsoids (M. Sullivan)

Using PMMA-PHSA particles suspended in polydimethylsiloxane (PDMS), it is possible to create both prolate and oblate ellipsoids by stretching or compressing the cross-linked PDMS matrix at temperatures above the particle glass transition temperature. From a processing standpoint, the challenge is separating ellipsoids from the PDMS without destroying them. We have developed a procedure to dissolve the elastomeric matrix surrounding the imbedded particles using an emulsion of 1.0 M tetrabutyl ammonium fluoride suspended in a solvent mixture composed of tetrahydrofuran and hexane. Following a two week period, the PDMS degrades to the extent that ellipsoids can be separated. We have successfully collected a small quantity for cleaning and size fractionation using a density gradient column technique. SEM images revealed a relatively high shape polydispersity but refining the stretching technique should help reduce this problem. Some preliminary depolarized dynamic light scattering experiments were preformed in order to study the rotational diffusion of the ellipsoidal latex particles suspended in organic solvents. Further refinement of the fabrication and characterization techniques is in progress.

E-Grid Cells (M. Sullivan)

The E-grid cell fabrication work has been optimized on a small scale. Low-defect electric grids can now be fabricated repeatedly. These grids can be made electrically active and move colloidal particles on surfaces by adjusting the frequency and voltage. We are improving the circuit fidelity by developing a procedure to properly stir the silver etchant during the etching step. The goal is to prevent Ag from precipitating on the electrode surfaces. The E-grid cells are being used for nucleation and growth studies similar to the static template cell experiments.

Microrheology (M. Sullivan)

In addition to active microrheology, e.g., manipulating a probe particle imbedded in a colloidal crystal using laser tweezers, we have started working on passive, i.e., thermally activated, microrheology of density matched colloidal suspensions. In particular, using the surface nucleated crystal growth technique, we are attempting to measure the cubic elastic constants of an FCC crystal. Analysis and experiments are underway.

Particle synthesis (A.D. Hollingsworth)

The preparation of low polydispersity PMMA-PHSA particles for Wes Kopacka’s research was completed during this period (see Fig. 1). Following the polymerization and subsequent processing, the particles appeared stable with no signs of aggregation. These steps included a solvent replacement cycle using dilute stabilizer solution to remove unreacted monomer in the diluent phase. This procedure allowed us to bypass the hexane distillation and thereby avoid the risk of secondary nucleation.

The final growth step required the functionalized stabilizer (R7698-13) to allow the subsequent covalent attachment of polymer chains to the particle surfaces. The stabilizer backbone molecules were ‘locked’ to the particle surfaces using an amine catalyst (N,N-dimethylethanolamine) at 120 deg. C for 2 hours. As the supply has diminished to practically zero, we will need to prepare and characterize more stabilizer for future particle syntheses.

SEM images of the suspension supernatant indicated that a minor fraction of below-average sized particles had been separated from the bulk population, thus narrowing the size distribution. The microscopy also revealed a small fraction of ‘fused doublets’ at a number concentration between 1000 to 2000 ppm. We believe these particles were formed during the locking reaction and may have been introduced from the residue that collected at the glass/nitrogen/suspension interface in the round bottom flask. The doublets appear to settle edgewise (enhanced sedimentation) and therefore could not be fractionated easily using centrifugation. Due to their shape and low number density, the doublets are not expected to have a significant affect on the planned light scattering experiments.

Particle size characterization using SEM and TEM indicated a discrepancy in microscope magnification which is being investigated presently. The XL-30 FEG-SEM has been recalibrated and images will be reanalyzed shortly. Samples were submitted to the FEI Company (formerly Philips) for microscopy using environmental SEM and sFEG-SEM microscopes. We will determine whether these techniques can reduce image artifacts, e.g., sample shrinkage, thereby improving image analysis.

Ongoing collaboration with Utrecht University (A.D. Hollingsworth, M. Leunissen)

We continue to study the new model system consisting of charge- and sterically stabilized PMMA particles in low-polar solvents (dielectric constant 5–8). Cyclohexyl bromide (CHB) and solvent mixtures with hexane and cis-decalin have been characterized in terms of conductivity to determine ionic strength; however, we have needed to make assumptions about the ions present. Generally, we believe the CHB molecule can decompose to form bromide ions and protons. This behavior can be induced thermally, or with irradiation (sonic or optical).

A recent idea was to transfer the CHB and its decomposition products to the water phase and study both pH and conductivity of the aqueous solutions as a function of time. The objective was to detect any systematic changes that might reveal the nature of the ions present in CHB and also tell us something about the decomposition mechanism. Our assumption has been that CHB can form free radicals that, by various propagation and termination reactions, can produce hydrobromic acid. HBr is a srong acid in water: pKa = –4.7 at 25 deg. C. If CHB (neat) contained this acid, one might expect that both the CHB as well as HBr would dissolve in water, provided we’re below the solubility limit. CHB is sparingly soluble in water: 640 microM at 18 deg. C.

Following careful sample preparation procedures, we measured conductivity as a function of CHB molarity. We found a time dependent, linear trend indicative of a strong electrolyte. When we plotted the proton (hydronium ion) concentration versus conductivity, all the data collapsed onto a single line with a slope consistent with the equivalent conductance of HBr (0.04294 S m2 /mol). The hydronium ion concentration also scaled linearly with time (initially) and indicated first order kinetics. The intercept values translated into an oil phase concentration of approximately 10 mM HBr. This value is consistent with Fuoss’s theory of ionic association using a value of 2 ? for the ionic radius (sum of the

radii of the cation and anion). Note that the oil phase ionic strength is of order 0.1 to 1.0 nM. For these experiments, the CHB conductivity (neat) was 1750 pS/cm.

After about 400 hours the 400 microM CHB saturated to [H+] = 260 microM; the 200 microM CHB plateaued to [H+] = 140 microM. These values suggest a significant amount of CHB decomposition in the aqueous phase. To verify the hypothesis, we will use silver nitrate as an indicator for bromide ions and perform a simple titration to determine the extent of decomposition. These tests should provide a solid basis for interpreting the organic solvent mixture conductivities as well as the suspension data. A paper covering our electrokinetic characterization studies is being written.

Princeton University Quarterly Report Progress: (05/1/04-07/31/04)

Static template cells (M. Sullivan)

Using static PMMA-PHSA templates produced in our lab, we have demonstrated the nucleation and growth of hard sphere crystals. Specifically, we have nucleated colloidal crystals from the FCC 100 face, which forces growth of a face-centered cubic structure. Using a magnetic wire and mixing magnet, it was possible to shear melt the crystal in situ. After mixing the sample, we observed growth from an initially disordered state to a finally ordered one. A series of confocal cubes was collected using a piezo-controlled, z-position stage on the Leica DM IRB inverted microscope. Video images were digitized and processed using image analysis methods we have developed. Extensive analysis is underway.

Preliminary results clearly show the growth of ordered regions as a function of time. The growth rate can thus be determined from this data. Measurement of the local volume fraction as a function of time has proven to be more challenging. The difficulty is due mainly to piezo stage errors, as well as the problems associated with tracking dense, disordered, fully-dyed particles. Only FCC 100 face static templates are currently available, but these exist in a variety of different lattice spacings. Future experiments could, for example, probe the effect of lattice commensurability as well as volume fraction dependence. With higher accuracy tracking, we hope to more clearly determine local volume fraction, and to observe volume fraction depletion at the crystal-liquid interface.

E-grid cells (M. Sullivan)

Electric grid cell construction continues to be refined, although most of the technical difficulties in producing these sample cells have been overcome. One remaining challenge is how to remove small shorts located within the interdigitated electrode pattern. One technique that works well when there are relatively few shorts is to pass a small current through the sample. Ideally, the cell would be non-conducting and would thus present infinite resistance. A ‘slightly’ shorted sample produces a large current at the points of contact which rapidly ablate under load. However, when there are many shorts in cell, this method causes large scale ablation, rendering the cell unusable. While the error rate is still high, it has been possible to produce electric grid cells with both 1-d patterns (lines) and 2-d patterns (FCC 100), the former being significantly easier. These cells have been used to demonstrate surface dielectrophoresis. We are using these for nucleation and growth studies similar to the static template cell experiments.

Preparation and characterization of colloidal ellipsoids (M. Sullivan)

We have successfully developed a method for producing ellipsoidal PMMA-PHSA particles in organic media. The ellipsoids are produced by first dispersing micron-size, spherical PMMA-PHSA particles in a mixture of polydimethylsiloxane (PDMS) resin and decalin. The PDMS prepolymer is crosslinked by heating the dispersion. The PMMA particles in the resultant elastomeric matrix are then heated above their glass transition temperature, followed by elongation at temperature, quenching, and finally selective chemical degradation the PDMS. Yields are low and the process is being refined, but ellipsoids of varying aspect ratios have been created in this way. SEM imaging revealed that the PMMA particles can withstand the chemical and physical treatments, and that the resulting particles are polydisperse in both shape and size. Nonetheless, stable, well-defined ellipsoidal particles were observed. A density gradient column is being developed to separate both small and aggregated particles from the suspension. The usefulness of photon correlation spectroscopy is being explored for the determination of size and shape of the nonspherical particles. The “hardness” of these particles remains to be tested. PCS+ Sample Preparation (A.D. Hollingsworth, C. Lant, K. Dendorfer)

Two engineering samples were prepared following the standard PMMA-PHSA particle synthesis. ADH012P1 particles were produced from a seeded growth polymerization (ADH012 seed diameter ~ 340 nm), whereas the ADH013 particles were synthesized in a single step reaction. Both polymerizations were carried out in a 100 ml round bottom flask. The suspensions corresponded to PCS+ Flight Sample 4 (crystalline, phi = 0.555) and PCS+ Flight Sample 7 (glass, phi = 0.625), respectively. From SEM image analysis of ADH012P1 (unswollen particles), we determined an average diameter of 0.51 microns with no detectable secondary nucleation. Using complementary dynamic and static light scattering, Wes Kopacka measured an average diameter of 0.525 microns and PDI of 0.039 for batch ADH013 (unswollen).

The lower volume fraction suspension appeared well index matched after the 2.5 hour solvent equilibration at 70 deg. C. Approximately 9.5 ml of the equilibrated sample (swollen diameter = 0.56 microns) was shipped in a glass vial to Zin Technologies for cell filling. Colloidal crystals were evident in both the vial and the flight cell. On May 11th, Chris Lant filled PHaSE cell #23 with this sample. The sample was easily transferred by syringe tubing, and there was no evidence of flow filtering. The new fill screw seal caused some pre-sealing leakage between the metal support ring plates, but there was no real problem otherwise. Upon inspection, however, Chris noted the sample in the cell was contaminated by many short fibers from unknown sources. Although Sample 4 was not ‘flight quality’ from this respect, the cell remained bubble-free. Fibers were also evident in the remaining sample in the vial.

The second flight cell, PCS-16, was filled on May 13th by Kevin with batch ADH013 (unequilibrated monodisperse PMMA/cis-trans decalin/tetralin). This sample was designed to become phi = 0.63 after 1 week at 50 deg. C. The fill and sealing went well, but the sample never seemed to reach the final volume fraction as evidenced by suboptimal index-matching. We observed similar behavior with the control sample (3 hour solvent equilibration at 70 deg. C). Batch ADH013 also exhibited unexpected swelling behavior in Wes Kopacka’s decalin-tetralin-tetrachloroethylene mixtures. The volumetric swelling factor for this particular batch may be slightly lower than the assumed value of 1.29; however, we have observed that PMMA-PHSA particle swelling in decalin-tetralin mixtures generally does not depend upon the particular batch or solvent lot used. ADH012P1 particles were dried from petroleum ether (high volatility solvent), whereas ADH013 was dried from n-dodecane for five days at 55 deg. C. It may be possible that ADH013

Task Progress:

particles were not completely dry when the index matching suspension was prepared, resulting in a lower than desired volume fraction.

PCS+ Flight System Testing (A. Bailey, A.D. Hollingsworth, B. Shiley, T. Lorik)

Flight hardware testing was conducted at Zin Technologies during the week of May 17th. Bill Shiley submitted a summary of PCS+ software issues identified during the PI testing and Art Bailey is preparing a detailed report.

Mirjam Leunissen Princeton Visit (May 29 through July 16, 2004)

A graduate student from van Blaaderen's lab (Utrecht University) was appointed Visiting Research Technical Staff Member I to continue her investigation of a new model system consisting of charge- and sterically stabilized PMMA particles in apolar solvents (dielectric constant 5–8). Cyclohexyl bromide (CHB) and solvent mixtures with hexane and cis-decalin were rigorously purified and characterized in terms of conductivity, refractive index, density and dynamic viscosity. Suspensions of fluorescently-labeled PMMA-PHSA spheres were prepared using these solvents.

Experiments included electrophoresis, dielectrophoresis ('electric bottle' cell) and a confocal microscopy study of phase behavior. The presence of water near the low density crystal appears to effectively 'deionize' the local organic phase as evidenced by unusually large depletion zones. Colloidosomes (stable hexagonal arrangements of particles on small water droplets dispersed in the oil phase) were also observed. A report detailing the results is being written.

Particle synthesis (A.D. Hollingsworth)

A process to synthesize relatively large batches of PMMA-PHSA particles in a controlled fashion was developed. Following this procedure, approximately 30 g (solid mass) of PMMA-PHSA particles were prepared for ongoing light scattering studies by Wes Kopacka. This quantity should be sufficient to fill eight PHaSE (PCS) cells at volume fractions ranging from 0.525 to 0.605. SEM image analysis indicated the average size diameter of these low polydispersity particles to be 0.60 microns. The particles were synthesized from batch ADH012 (2R = 370 nm) via a three-stage, seeded-growth polymerization.

The process development results will be utilized in the preparation of PCS+ flight samples. An important discovery was that the commercial stabilizer X190-442 (ICI) can be used to stabilize particles during the growth steps. As demonstrated previously, X190-442 cannot be used for single-step polymerizations which may be due to its broad molecular weight distribution and/or higher Tg (due to the absence of glycidyl methacrylate in the PMMA backbone). The epoxide groups required for chemical attachment of stabilizer backbone to particle surfaces were provided by the R7698-13 stabilizer used in the final growth step. Secondary nucleation was observed occasionally; however, we believe that this phenomenon is related to the reactor scale-up and not the type of stabilizer used. Small particles appear to be nucleated when the monomer feed is too concentrated or the feed rate is too rapid. This behavior occurred using either stabilizer. Secondary particle nucleation is usually accompanied by irreversible particle aggregation. We found that the small particles could not be separated from primary ones. From SEM images, they appeared to permanently decorate the larger population, thus suppressing/preventing their crystallization.

Future work will include the synthesis and characterization of a PHSA comb-graft polymeric stabilizer to replenish the R7698-13 batch. The monomer-particle size calibration curve will be redetermined for the 0.1 and 0.5 liter scale reactors.

Princeton University Quarterly Report Progress: (02/1/04-04/30/04)

E-grid cells (M. Sullivan)

Several fabrication issues have been addressed including the simultaneous printing of micro- and macro-scale circuit features. The FCC 100 forcing pattern (square lattice) was replicated successfully on 24 x 60 mm, no. 1-1/2 cover slips (borosilicate glass) using the soft lithography process. Using a wet grinding wheel at the glass shop (Chemistry Department), we cut 45 deg. bevels on the long sides of (10) standard microscope slides with relative ease. With cover slips attached to the modified slides, a channel will be created in which the fine wire can be connected and secured. Prototype cells were wired, filled with sample, sealed and energized for confocal microscopy observation. To produce an 'anti-stick' surface, the interior surfaces (silver electrodes and glass) were coated with a dilute solution of poly(methyl methacrylate-co-methacrylic acid) resin followed by the physical adsorption of the PHSA graft copolymer stabilizer. Although the PMMA-PHSA particles were undersized for this particular circuit pattern, we were able to manipulate particles at the surface layer in a controlled manner and did not observe particle sticking over a period of several days. Pro-E cell design drawings (no. SK-10585 through SK-10589) for the E-field and E-Grid cells were produced by NGIT.

Static template cells (M. Sullivan)

Static surface templates have been used to induce patterned heterogeneous nucleation. The e-beam etched pattern (silicon wafer) was reproduced successfully on the PMMA coated cover glass using the PDMS stamping technique. The PMMA pattern was then treated with PHSA stabilizer to produce an 'anti-stick' surface with the desired template morphology. The experiment consisted of concentrated suspensions (phi > 0.49) of monodisperse PMMA-PHSA spheres in a density matching solvent mixture of decalin and tetrachloroethylene. These were particles synthesized at Princeton (batch ADH001) with a swollen diameter of approximately 1.2 microns and were rhodamine 6G labeled. Unpatterned surfaces nucleated hexagonally packed faces of RHCP crystals, whereas surfaces patterned with two-dimensional square lattices can nucleate and grow a variety of structures depending on the lattice spacing. In particular, a square lattice spacing commensurate with the bulk lattice spacing grew FCC (100) crystals well away (> 40 microns) from the surface. 3-D particle tracking was accomplished by stitching 2-D confocal slices to form the 3-D image. Although the z-resolution is not as good as the Mischa Megans/Tony Apicella algorithm, the fully fluorescent particles can be located more accurately.

Particle synthesis (A. Hollingsworth)

Most of the focus was related to preparation of engineering samples for the upcoming PCS+ flight hardware testing (5/17 through 5/21/04). Reactor scale-up experiments were conducted determine the process conditions necessary to produce large sample quantities. So far, it appears that the monomer-particle size calibration established for the 100 ml reactor does not translate to the ½ liter vessel. We carried out two single stage polymerizations (batch ADH011 and 012) in a 500 ml round bottom flask (RBF) with the same materials used to produce monodisperse PMMA spheres in the 100 ml

scale process. Following the 2.5 hour reaction, particle masses were 35 and 50 g, respectively, about 3 to 5 times the small scale yield. The material balances indicated complete monomer conversion; however, SEM image analysis of the particles showed that the number average diameter was approximately 340 nm in both cases. This result was significantly smaller than the 600 nm target size, although polydispersity was low and colloidal crystals were observed in the sedimented layers for both samples.

It appears that heat transfer between the reactor and the oil bath is significantly different for the two scales. A measurement of reactor temperature during the heating stage revealed a longer warm-up period for the 500 ml RBF followed by temperature overshoot (above the 80 deg. C set point). Particle nucleation occurs during this relatively short period and it is known that reactor temperature correlates with particle size (see for example, Croucher & Winnik in An Introduction to Polymer Colloids (1990)). PID controller tuning will probably solve this problem.

Although the average particle sizes produced are smaller than desired, seeded growth polymerization was used to achieve larger spheres in a controlled manner. A single stage growth reaction in the 100 ml RBF using ADH012 seeds produced the desired 520 nm diameter particles with no detectable secondary nucleation. A scale up to the 500 ml RBF was unsuccessful, however, producing a second population of small particles which proved difficult to separate from the 600 nm diameter primary particles. In this experiment, the monomer addition rate was too fast causing a gradual buildup of methyl methacrylate and methacrylic acid (comonomers) during the polymerization. Slower addition was not possible due to hexane evaporation over the 10 hour feeding period, so a two-step growth experiment is planned. Extensive characterization of the processed suspension (SEM, dynamic and static light scattering) is underway.

The engineering samples were formulated from ADH012P1 (seeded growth) and ADH013 (single stage) as PCS+ flight sample numbers 4 and 7 (phi = 0.555 and 0.625, respectively). Using complementary dynamic and static light scattering, Wes Kopacka determined an average diameter of 525 nm and PDI of 0.039 for batch ADH013. The lower volume fraction suspension appeared well index matched after the 2.5 hour solvent equilibration at 70 deg. C. Batch ADH013 exhibited unexpected swelling characteristics in both the decalin-tetralin and decalin-tetralin-tetrachloroethylene mixtures. The equilibrated sample was not as well matched indicating that the assumed volumetric swelling factor for this particular batch is slightly lower than the assumed value of 1.29. Flight cells were filled with Samples 4 (fully equilibrated) and 7 (unequilibrated); the latter one was heat soaked at 50 deg. C for 3.5 days to a point where the suspension appeared closely matched, but not fully equilibrated.

Future work (May-June-July period) includes the production of more PMMA particles for Wes Kopacka's research via the two-stage seeded growth of ADH011; testing the ICI nonaqueous dispersant (X190-442) as a stabilizer for the seeded growth polymerizations; synthesis and characterization of a PHSA comb-graft polymeric stabilizer; and the development of IDL software to analyze SEM images of PMMA particle monolayers (number average size and polydispersity).

Specialty Cell Interface Design Review Meeting 2/6/04

M.T. Sullivan and A.D. Hollingsworth participated in the telecon. Summary and action items:

Mechanical interfaces: Changes to the current designs for the E-grid and E-field cells will include: shorter cover slips to accommodate mechanical and electrical interfaces; the bevel for the electrical contacts will be 0.75 mm with a 45 deg. angle; the Princeton team will investigate using either gold or silver for the E-grid cell template, but not both.

Electrical interfaces: The E-grid and E-field cells have different voltage requirements and will therefore not reside on the same platen. The cell-to-platen, platen-to-stage, stage-to-control circuit electrical interfaces are undefined. The platen-to ETM and ETM-to STES electrical interfaces are undefined.

Power requirements: The E-field cell module will require continuous power for at least one week. Power interruptions (during a transfer from LMM to STES) of up to one hour are acceptable. The E-grid cell module will require power continuously for up to 24 hours. Concerns were raised regarding transfer of modules to the STES from a power perspective (power loss for an undefined period), a control perspective (how to maintain the same operating conditions (voltage, frequency) during and after transfer, and a microgravity perspective (what level of disturbance can be tolerated by the cells during transfer). The current Specialty Cell Control circuit resides in the LMM Control Box and provides a variable frequency (square wave) from approximately DC to 100 kHz at a fixed 30 V peak to peak. This voltage does not meet the desired requirements. Action for Eric Miller and John Zoldak: estimate cost and schedule impact to design power control for LMM and STES. Due: 02/13/04.

Operating Parameters (number of circuits, power durations): There will be 10 modules for each cell design. Each platen will be populated with one type and they will operate on one circuit. Experimental runs will consist of 2 to 3 scenarios per platen. The stir bar length causes possible short circuiting of the E-field cell and may create sample homogenization problems. Action for the Princeton team: solve the short circuit issue. Due: 04/15/04. Action for Rick Rogers and Bill Meyer: determine the best homogenization regimen given the system that has already been designed. Due: 02/20/04.

Imaging Techniques: Confocal microscopy will be used primarily, but the NGIT team needs to determine the best way to locate the E-grid within the microwell. R. Rogers and M. Sullivan will investigate techniques for determining the crystal alignment. Action for T. Bruzas: send the Princeton Team four cell bases, as well as epoxy and the cover slip attachment protocol. Due: 02/13/04. Action for T. Bruzas: contact M. Wirks and obtain the verification plan for STES interfaces. Due: 02/13/04. Action for A. Hollingsworth: send high volume fraction, core shell particles to R. Rogers within the next three business days.

Q-Imaging camera

The camera received from NASA GRC works well for single image capture. Recording to video isn't possible yet but we are investigating the required software.

PMMA microspheroids (Burton, Sullivan)

This project was initiated as a Junior Paper by a Princeton undergraduate majoring in physics. We are attempting to produce monodisperse spheroidal particles from PMMA-PHSA spheres following the process developed by Keville et al. (1990). One micron diameter, rhodamine labeled PMMA spheres were suspended in PDMS elastomer which was then cured. The composite elastic matrix was deformed under uniaxial tension at a temperature well above Tg of the imbedded particles to produce the microspheroids. Then the material was quenched and the PDMS was dissolved by selective chemical degradation. The initial results show that we can reproduce the Keville experiment; however, the

particle stabilizer is different. Particle yield is relatively low.

Princeton University Quarterly Report Progress: (11/1/03-01/31/04)

E-grid and E-field cells (M. Sullivan)

Using the redesigned master wafers prepared at Cornell University's CNL last fall, several electric grid circuits were printed on no. 1-1/2 cover glass via the soft lithography process developed here. The new designs provide multiple patterns wired as a single electrical circuit and can be contained within the 2.5-mm diameter microwell cell. For example, one design features twenty-five fields (128 microns x 128 microns each) arranged in a five by five array. Patterns including FCC and HCP forcing planes, as well as hexagonal planes, all with a variety of lattice spacings, are located in the active region. The new patterns can be faithfully reproduced in silver; however the fidelity of the gold electrodes is not as good. This discrepancy may be related to differences in the chemical etching rate. Although the silver patterns appear acceptable at 100x magnification, defects in the pattern replication are still preventing repeatable electrically connectivity. Work is being done to refine the procedure and create electrically isolated, interdigitated electrodes.

Following the PDMS stamping, we always find some residual electrical contact at the edge of the stamped pattern. The clean up procedure involves scratching the metal regions to remove electrical contact. If the cell remains electrically connected, the next step is to look for obvious conductive defects in the stamped electrode pattern. Here, metal defects may be observed between adjacent digits of the interdigitated electrode pattern. They can be removed by "burning" the offending connection using the laser tweezers. After removing all visible defects, the resistance usually rises significantly. If this does not occur, the pattern was not etched correctly and is discarded. If a large, but finite, resistance persists, electrical current can be passed through the pattern using a power supply to ablate any remaining conductive patches. After this procedure, the resistance is usually infinite and the pattern remains intact.

The simpler E-field cell is used to change and control in situ particle concentration. An electric field is created using two metal electrodes located on the outer surface of the glass cell. This configuration creates a thin, finite width, parallel plate capacitor. When the cell is energized, a large uniform electric field develops at the center and field fringing produces a steep field gradient at the edge of the electrodes. A particle concentration gradient develops with time, the osmotic pressure of which balances the dielectrophoretic force. As a result, the dielectrophoretic force pumps particles from the low field regions outside the capacitor to the high field regions inside of it. The density of the particles in the approximately force free region between the plates is readily adjustable by setting the capacitor voltage thus allowing control of volume fraction in colloidal suspensions.

The open question has been whether it is possible to achieve a glassy state using E-field cells. The main concern is that the E-field densification proceeds significantly slower than the crystal nucleation, so the sample would tend to crystallize before densifying at glassy volume fractions. By carefully shear melting in the dense central region while avoiding mixing the central and outer regions; the interior region can be disordered while maintaining the density gradient. Using this technique, we have been able to introduce disorder and create "glassy" samples using an E-field cell.

Together with Tony Bruzas at Northrop Grumman (NGIT), we are finalizing E-field and E-Grid cell designs and are creating Pro-E drawings. The design drawings will be submitted to NGIT on or before the interface design review scheduled for February 19, 2004 at NGIT. Several jpg files of the rough drawings have been uploaded to the NASA FTP file server. A key feature is the electrode connection on the E-Field cells. The upper electrodes are on the coverslip and are connected in the same way as the E-grid cell. The lower electrodes are on the specialty base itself. These electrodes are sputtered at a 45-degree angle to allow the deposited metal to connect around corners. We have demonstrated this with a standard coverslip and shown that the technique works well. That is, we can sputter an electrical contact fully around the edge of the thin glass. The electrode is also sputtered down the beveled edge so as to connect with the silver-filled epoxy.

Static template cells (M. Sullivan)

We have reproduced e-beam etched patterns on to PMMA coated cover glass using a commercial poly(methyl methacrylate-co-methacrylic acid) resin. The PMMA patterns are then treated with PHSA to produce an "anti-stick coating" with the desired template morphology. The acid functionality of the copolymer allows the chemical attachment of the PHSA backbone polymer using 2-dimethylaminoethanol catalyst at 120 deg. C. We have used the static surface templates to induce patterned heterogeneous nucleation in hard sphere crystals. The experiment consisted of concentrated suspensions (phi > 0.49) of monodisperse PMMA-PHSA spheres in a density matching solvent mixture of decalin and tetrachloroethylene. Unpatterned surfaces nucleate hexagonally packed faces of RHCP crystals, whereas surfaces patterned with two-dimensional square lattices nucleate and grow a variety of structures depending on the lattice spacing. In particular, a square lattice spacing commensurate with the bulk lattice spacing grows FCC (100) crystals well away (> 40 microns) from the surface.

Particle synthesis (A. Hollingsworth)

We performed a series of seeded-growth polymerizations to produce the first rhodamine 6G labeled, core-shell PMMA particle suspension. A batch of ~370 nm dia. PMMA-PHSA cores (designation ADH003) were used as seed particles and grown to approximately 1.04 micron diameter in three stages. The fluorescent dye was incorporated in the core particles using an acetone-cyclohexanone solution. SEM images showed evidence of minor secondary nucleation; however, the small particles (< 100 nm diameter) could be easily separated by centrifugation following each polymerization. As expected, the particle size distribution appeared to narrow significantly with increasing size.

Because the methyl methacrylate and methacrylic acid monomers are absorbed into the seed particles during the growth stages, the dye tended to spread out of the core region. Nonetheless, confocal microscopy indicated that the particles in close packing were not homogeneously fluorescent, i.e., they appeared to be brighter in the middle. The outermost shell appeared to be nonfluorescent, which should improve particle tracking in dense suspensions. Because the integration times were longer using the Leica camera (as compared with fully dyed particle samples), we used unattenuated laser power to image the particles. They did not appear to photobleach significantly during the observation time. Another obvious difference between these particles and the homogeneously fluorescent PMMA spheres was that the final stage appeared nearly white with a low level of fluorescence. Fully dyed particles in decahydronaphthalene (decalin) produce an opaque, pink-orange color. NGIT observed similar results with the core-shell PMMA particle suspension provided to them in December.

The core-dyed particles are expected to be stable with respect to dye migration provided they remain suspended in a poor solvent such as decalin. We plan to test our centroid tracking algorithms on these particles to determine whether there's any improvement over fully dyed PMMA spheres. For the next phase of core-shell particle development, we will prepare (or possibly obtain) a rhodamine 6G dye monomer. The essential features of the particle synthesis will be (1), the copolymerization of this monomer with methyl methacrylate and methacrylic acid; (2), the simultaneous cross linking reaction using ethylene glycol dimethacrylate (EGDM); and (3), particle swelling to wash out low molecular weight dye molecules (monomers, oligomers, etc.). A detailed schedule for sample delivery is being prepared.

We received Hua Hu's preprint, "Preparation of fluorescent particles with long excitation and emission wavelengths dispersible in organic solvents", co-authored with Ron Larson at the University of Michigan. According to the paper, they have successfully synthesized a PHSA stabilizer similar to that described by Antl et al. (1986). The important modifications are the recrystallization of the 12-hydroxystearic acid (12-HSA) and the elimination of glycidyl methacrylate monomer in the third stage, i.e., no pendant epoxide groups from the backbone for the subsequent 'locking reaction'. By avoiding the chemical attachment of the stabilizer to the particle surface, Hu claims to have reduced polydispersity and avoids thermal degradation of the DiIC18(5) dye. We believe the locking reaction is necessary to suspend our particles in index matching solvents (which tend to swell the particles), and to prevent washing off physically adsorbed stabilizer during the particle cleaning procedure.

An important point made in the paper is that the 12-HSA should have an optimal purity around 85 to 90%. The impurity is believed to be stearic acid, which reduces the degree of polycondensation through termination reactions. In other words, to achieve the desired 5 to 6 ester linkages per PHSA chain would require the reaction of 1 stearic acid molecule for every five to six 12-HSA molecules. If the acid purity is higher than 90%, the degree of polymerization will be higher, which will produce longer stabilizer chains on the particle surface. The polymer molecular weight distribution may also broaden. Generally, we expect that PMMA particles would tend to be less stable and more polydisperse using the suboptimal stabilizer. This hypothesis may explain the difficulties encountered by van Blaaderen, et al. (Utrecht) and Schofield (U. Edinburgh). If the purity of 12-HSA was less than 85%, the degree of polymerization would be lower, rendering the stabilizer ineffective. Ideally, the MW should be ~ 1700 to produce ~ 9 nm (fully stretched) "teeth".

To synthesize additional PHSA stabilizer required for PHaSE-2 as well as PCS+ sample preparation, we will blend 99% purity 12-HSA with 99+% stearic acid in a 6:1 molar ratio and then polyesterify the 12-HSA side chains to the desired molecular weight. The second stage involves reacting the terminal carboxylic acid with an equimolar amount of glycidyl methacrylate to produce a terminal methacrylate double bond; finally, a mixture of the PHSA-GMa adduct macromonomer (50% in hydrocarbon solution) with methylmethacrylate and glycidyl methacrylate is polymerized to produce the desired comb-graft copolymer. The theoretical weight ratio of the monomers will be:
PHSA-GMa/MMa/GMa = 50/45/5. This procedure should result in a final stabilizer product similar to the 'good' (Julian Waters) stabilizer, now down to ~25 ml, e.g., about twenty-five 20g batches at 40% w/w solids. We will also try the commercial Uniqema Hypermer LP1 polycondensed fatty acid product beginning with Stage II.

The particle synthesis lab was successfully moved to the Colloidal Physics Lab in Bowen Hall. It had been set up in the Russel lab in the Engineering Quadrangle since late 2001. The move will improve the efficiency of particle synthesis development and sample production by "having everything under one roof" and was prompted by upcoming lab renovations. Using the relocated facility, we very quickly and successfully prepared a batch of 750 nm diameter PMMA-PHSA particles (designation ADH010) for patterned heterogeneous nucleation experiments. The monomer concentration-particle size calibration is based on the R7698/13 (J. Waters) stabilizer and allowed us to prepare low polydispersity particles at the desired size.

PCS mixing tests (A. Bailey, A. Hollingsworth)

At this time, we are addressing the following open questions: (1) How to fill the high volume fraction samples (i.e., phi = 0.600, phi = 0.625); (2), which sample cells require fins? (are fins necessary to adequately mix the samples?); (3), do we understand the particle swelling behavior (tetralin-PMMA equilibration) at lower temperatures, i.e., 45 to 50 deg. C. Bragg scattering data from the various PCS cells is being analyzed now. Regarding the seven PCS+ flight samples, Princeton will provide one or two volume fractions by April 2004. During the next two months we will synthesize, characterize, clean and resuspend particles to produce approximately 8 to 10 ml of suspension per sample at volume fractions ranging from 0.505 to 0.625. The present polymerization system has been scaled up by a factor of five and there is a sufficient amount of polymeric stabilizer to produce the flight samples.

NASA visit to Princeton University November 13, 2003

Members of the LMM team (Bill Meyer, DeVon Griffin, Bhim Singh) visited the Chaikin-Russel lab at Princeton University last November to discuss specialty cell development, sample production and delivery schedules. Specific questions related to the Science Requirements Document for PHaSE-2 were also discussed.

LMM Critical Design Review meeting December 10–11, 2003

A. Hollingsworth represented Princeton University at the CDR meeting at NGIT last December. P. Chaikin participated by telephone. Among the concerns raised by NASA project scientists for CDR review panel attention were issues regarding high volume fraction cell filling and leak integrity, and microscope optical resolution performance under the vibration conditions of LMM. Other issues included synchronization of the camera to the confocal unit, long equilibration times needed for E-field experiments, and multiple cell mixing by the sample homogenization system.

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Abstracts for Journals and Proceedings	Zhao, K.; Sullivan, M.T. (Princeton University); Harrison, C.K. (Schlumberger-Doll Research Center); Russel, W.B.; Chaikin, P.M. (Princeton University) "Properties and Manipulation of Square Colloidal Platelets in Light and Electric Fields" American Physical Society March Meeting, Montreal, Canada Session P22 - Colloids & Emulsions: Statics. ORAL session, Wednesday midday, March 24 Volume 49, No. 1; p. 935 , Mar-2004
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